RACE Regular Analysis Through Continuous Exercise

States of Matter : RACE-01

Ideal Gas Laws: Boyle's Law

1. A volume of 2.5 L of a sample of a gas at 27°C and 1 bar pressure is compressed to a volume of 500 ml keeping the temperature constant, the percentage increase in pressure is

(A) 100% (B) 400% (C) 500% (D) 80%

CSM185

Charles' Law

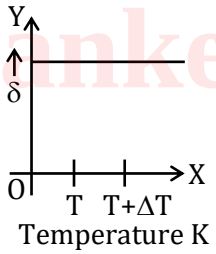
2. A sample of gas at 35°C and 1 atmospheric pressure occupies a volume of 3.75-L. At what temperature should the gas be kept, if it is required to reduce the volume to 3.0 L at the same pressure?

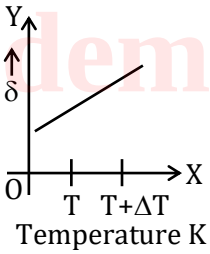
(A) −26.6°C (B) 0°C (C) 3.98°C (D) 28°C

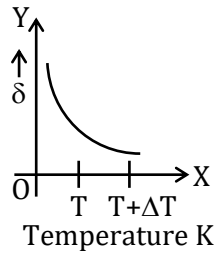
CSM187

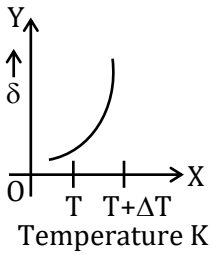
3. An ideal gas is initially at temperature T and volume V. Its volume increases by ΔV due to an increase in temperature of ΔT, pressure remaining constant. The quantity $\delta = \frac{\Delta V}{V\Delta T}$ varies with temperature as

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(A)


(B)


(C)


(D)


CSM236

Gay Lussac's Law

4. If the pressure of a certain amount of a gas increases by 1% on heating by 1°C at constant volume, then its initial temperature must be

(A) 100 K (B) 100°C (C) 250 K (D) 250°C

CSM188

5. A vessel contains 1 mole of O_2 gas at a temperature T. The pressure of the gas is P. An identical vessel containing 1 mole of He gas at a temperature 2T has a pressure of

(A) P/8 (B) P (C) 2P (D) 8P

CSM193

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Avogadro's Law

6. The diameter of a balloon filled by 8 moles of He gas is 10 cm. If 7 moles of He gas effuses out in the night, then what should be the diameter of balloon in the next morning? Assume constant pressure and temperature for the gas.

(A) 5 cm (B) $\left(\frac{7}{8}\right)^{1/3} \times 5\text{cm}$ (C) $\left(\frac{8}{7}\right)^{1/3} \times 5\text{cm}$ (D) 20 cm

CSM240

Ideal Gas equation; Graphs of Ideal Gas

7. The molar volume of CO_2 is maximum at

(A) 273 K and 1 atm (B) 546 K and 1 atm (C) 273 K and 2 atm (D) 546 K and 2 atm

CSM192

8. A bulb with chlorine gas at ambient pressure contains 3.5 g at t°C. When the bulb is placed in a second thermostat at a temperature 30°C higher and the stopcock of the bulb temporarily opened (and closed again) to restore the initial pressure, the bulb is now found to contain 3.2-g of chlorine. What is the value of t°C ?

(A) 47K (B) 14.1°C (C) 320°C (D) 47°C

CSM194

9. Pressure of 1 g of an ideal gas A at 27°C is found to be 2 bar. When 2 g of another ideal gas B is introduced in the same flask at same temperature the pressure becomes 3 bar. What is the relationship between their molecular masses?

(A) $2M_B = M_A$ (B) $M_B = M_A$ (C) $M_B = 4M_A$ (D) $M_B = 2M_A$

CSM196

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10. A spherical air bubble is rising from the depth of a lake where pressure is P atm and temperature is T K. The percentage increase in its radius when it comes to the surface of a lake will be (Assume temperature and pressure at the surface to be respectively 2T K and P/4)

(A) 100% (B) 50% (C) 40% (D) 200%

CSM252

Dalton's Law of Partial Pressure

11. Which of the following mixture of gases at room temperature follows Dalton's law of partial pressure?

(A) NH_3, HCl (B) H_2, O_2 (C) NO, O_2 (D) All of these

CSM199

12. Assume that air is 21% of O_2 and 79% of N_2 by volume. If the barometric pressure is 740 mm, then the partial pressure of O_2 is closest to which one of the following?

(A) 155 mm (B) 310 mm (C) 580 mm (D) 740 mm

CSM200

Vapour Pressure

13. A vessel has N_2 gas saturated with water vapours at a total pressure of 1 atm. The vapour pressure of water is 0.3 atm. The contents of this vessel are completely transferred to another vessel having one-third of the capacity of original volume, at the same temperature. The total pressure of this system in the new vessel is

(A) 3.0 atm (B) 1 atm (C) 3.33 atm (D) 2.4 atm

CSM206

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14. A vessel of 5.0 L capacity contains 1.4 g nitrogen at 1800 K. Assuming that at this temperature, 40% of molecules are dissociated into atoms and the gas is ideal, what is the gas pressure?

(A) 2.07 atm (B) 1.476 atm (C) 1.05 atm (D) 2.67 atm

CSM258

15. If saturated vapours are compressed slowly and isothermally to half of the initial volume, then the pressure exerted by the vapours will

(A) become four times (B) become double

(C) remain unchanged (D) become half

CSM266

16. Vapour is injected at a uniform rate in a closed vessel which was initially evacuated. The pressure in the vessel

(A) increases continuously. (B) decreases continuously.

(C) first increases and then decreases. (D) first increases and then becomes constant.

CSM267

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	A	C	A	C	A	B	D	C	A
Question	11	12	13	14	15	16				
Answer	B	A	D	A	C	D				

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States of Matter : RACE-02

Container Problems

1. A perfectly expandable balloon filled with helium gas at 27°C and a pressure of 720 mm of Hg has a volume of 100 L. The balloon rises to an altitude where the pressure is 420 mm of Hg and the temperature –63°C. What is the change in the volume of the balloon?

(A) 16 L (B) 20 L (C) 48 L (D) 120 L

CSM197

2. A closed vessel contains equal number of nitrogen and oxygen molecules at a pressure of P mm. If nitrogen is removed from the system, then the pressure will be

(A) P mm (B) 2P mm (C) P/2 mm (D) P^2 mm

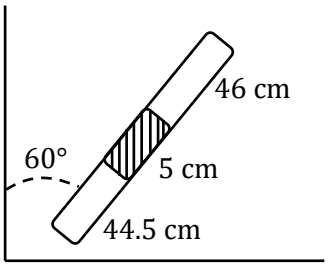
CSM202

3. The volumes of the two vessels are in the ratio of 2 : 1. One contains nitrogen and the other oxygen at 800 mm and 680 mm pressure, respectively. Determine the resulting pressure when they are connected together.

(A) 760 mm (B) 670 mm (C) 1140 mm (D) 1480 mm

CSM204

4. A thin tube of uniform cross section has trapped air columns of lengths 46 and 44.5 cm above and below a middle pellet of mercury 5 cm in length (see figure) when held at an angle of 60° to the vertical. When placed horizontally, the columns are equal in length. The temperature is 27°C. The pressure of air columns, in cm Hg, when the tube is horizontal, is



(A) 76.0 cm (B) 71.7 cm (C) 73.2 cm (D) 75.4 cm

CSM231

Graham's Law of Diffusion and Effusion

5. A quantity of 4 g H_2 reacts with 9×10^{23} Cl_2 molecules and forms HCl gas. If the total pressure of the system after reaction is 700 mm, then what is the partial pressure of HCl? Assume complete reaction. ($N_A = 6 \times 10^{23}$)

(A) 200 mm (B) 350 mm (C) 300 mm (D) 600 mm

CSM205

6. If helium and methane are allowed to diffuse out of the container under similar conditions of temperature and pressure, then the ratio of rate of diffusion of helium to methane is

(A) 2 : 1 (B) 1 : 1 (C) 1 : 2 (D) 4 : 1

CSM210

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7. A gas with formula C_nH_{2n+2} diffuses through the porous plug at a rate one-sixth of the rate of diffusion of hydrogen gas under similar conditions.
The formula of gas is
(A) C_2H_6 (B) $C_{10}H_{22}$ (C) C_5H_{12} (D) C_6H_{14}

CSM212

8. The ratio of universal gas constant and molar mass of gas is called molar gas constant. The value of molar gas constant is greater for
(A) He (B) N_2 (C) H_2 (D) Same for all

CSM254

9. The rate of diffusion of two gases X and Y is in the ratio 1 : 5 and that of Y and Z is in the ratio of 1 : 6. The ratio of the rate of diffusion of Z with respect to X is
(A) 30/1 (B) 1/30 (C) 5/6 (D) 6/5

CSM269

10. At 20°C, two balloons of equal volumes and porosity are filled to a pressure of 2 atm, one with 14 kg N_2 and other with 1 kg of H_2. The N_2 balloon leaks to a pressure of half atm in 1 hr. How long will it take for H_2 balloon to reach a pressure of half atm?
(A) 1 hour (B) 30 minutes (C) 16 minutes (D) 3.74 hours

CSM271

11. A mixture of CH_4 and HBr in a vessel is allowed to effuse out through a small hole at the same temperature. What is the mole fraction of CH_4 if the initial rates of effusion are the same for both gases?
(A) 0.31 (B) 0.44 (C) 0.5 (D) 0.16

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CSM273

Kinetic Molecular Theory of Gases (KTG)

12. The RMS speed of oxygen molecules in a gas is V. If the absolute temperature is doubled and the oxygen molecules are dissociated into oxygen atoms, then the RMS speed will become
(A) V (B) $\sqrt{2}$ V (C) 2V (D) 4V

CSM277

13. For two gases, A and B with molecular weights M_A and M_B, it is observed that at a certain temperature T, the mean velocity of A is equal to the root mean square velocity of B. Thus, the mean velocity of A can be made equal to the mean velocity of B, if
(A) A is at temperature T, and B at T', T-> T'.
(B) A is lowered to a temperature T' < T while B is at T.
(C) both A and B are raised to a higher temperature.
(D) both A and B are placed at lower temperature.

CSM282

14. The most probable kinetic energy of gas molecule is
(A) kT/2 (B) 3kT/2 (C) kT (D) kT/4

CSM297

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Kinetic Energy and Molecular Speeds; Maxwell's distribution of molecular speed

15. Three closed vessels A, B and C are at the same temperature and contain gases which obey the Maxwellian distribution of velocities. Vessel A contains only O_2, B only N_2, and C, a mixture of equal quantities of O_2 and N_2. If the average speed of the O_2 molecules in vessel A is V_1, then that of the N_2 molecules in vessel B is V_2, the average speed of the O_2 molecules in vessel C is

(A) $(V_1+V_2)/2$ (B) V_1 (C) $\sqrt{V_1V_2}$ (D) $\sqrt{3kT/M}$

Here, M is the mean molar mass of the mixture

CSM278

16. When temperature is increased, the difference between most probable velocity, RMS velocity and average velocity

(A) increases (B) decreases (C) remains the same (D) None of these

CSM279

17. At what temperature does the average translational K.E. of a molecule in a gas becomes equal to the K.E. of an electron accelerated through a potential density of 3 V ?

(A) 232 K (B) 2320 K (C) 23,200 K (D) 2,32,000 K

CSM287

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	C	A	D	D	A	C	C	A	C
Question	11	12	13	14	15	16	17			
Answer	A	C	B	A	B	A	C			

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RACE
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States of Matter : RACE-03

Collision theory: Z_1, Z_{11}, Z_{12}, Mean free path

1. If X is the total number of collisions which a gas molecule registers with other molecules per unit time under particular conditions, then the collision frequency of the gas containing N molecules per unit volume is

(A) X/N (B) NX (C) 2NX (D) NX/2

CSM219

2. Which of the following gas molecule has the longest mean free path at the same pressure and temperature?

(A) H_2 (B) N_2 (C) O_2 (D) Cl_2

CSM220

3. The number of gas molecules effusing per second unit area through an orifice on the wall of the container is $\frac{1}{4}$. $N^*.u_{av}$, where N^* is the number of molecules per unit volume. What is this number under STP conditions for nitrogen gas?

(A) $1.53 \times 10^{24}\ m^{-2}s^{-1}$ (B) $1.03 \times 10^{22}\ m^{-2}s^{-1}$ (C) $3.05 \times 10^{27}\ m^{-2}s^{-1}$ (D) $3.05 \times 1021\ m^{-2}s^{-1}$

CSM283

4. The number of collisions of Ar atoms with the walls of container per unit time, keeping the partial pressure of Ar constant,

(A) increases when the temperature increases.

(B) remains the same when CO_2 is added to the container at constant temperature.

(C) increases when CO_2 is added to the container at constant temperature.

(D) decreases when the average kinetic energy per molecule is decreased.

CSM289

5. How many times the average speed of the molecules in a gas becomes when the temperature is raised from 27°C to 159°C?

(A) 1.2 (B) 1.44 (C) 5.89 (D) 2.43

CSM291

6. When the temperature of an ideal gas is increased at constant pressure,

(A) The collision number increases. (B) The collision frequency increases.

(C) The mean free path increases. (D) The number of molecules per unit volume increases.

CSM293

Behaviour of Real Gases; Deviation from Ideal Gas Behaviour; Introduction of Van Der Waal's equation

7. The van der Waals equation for (½) mole of a gas is

(A) $\left(P+\frac{a}{V^2}\right)(V-b)=RT$ (B) $\left(P+\frac{a}{4V^2}\right)\left(V-\frac{b}{2}\right)=\frac{RT}{2}$

(C) $\left(P+\frac{a}{4V^2}\right)\left(\frac{V-b}{2}\right)=RT$ (D) $\left(P+\frac{a}{4V^2}\right)\left(\frac{V-b}{2}\right)=2RT$

CSM222

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8. What is the pressure exerted by 32 g of methane in a 250 ml vessel at 300 K using van der Waals equation. Given that a = 2.5 atm L mol^{-2} and b = 0.05 L mol^{-1}.

(A) 227.54 atm (B) 299.64 atm (C) 168.4 atm (D) 328.4 atm

CSM223

Compressibility Factor ; Graphs of Compressibility Factor

9. Above Boyle's temperature, a gas

(A) may be liquefied by suitable pressure.

(B) shows positive deviation at all pressure.

(C) shows negative deviation at low pressure and positive deviation at high pressure.

(D) shows positive deviation at low pressure and the deviation continuously increases with increase in temperature.

CSM314

10. The expression for compressibility factor for one mole of a van der Waals gas at Boyle temperature is

(A) $1+\frac{b^2}{V(V-b)}$ (B) $1+\frac{b^2}{V^2}$ (C) $1+\frac{b}{V}$ (D) $1-\frac{b^2}{V^2}$

CSM315

Problems on Real Gas

11. Under what pressure will carbon dioxide have the density of 2.2 g/litre at 300 K? For CO_2, a = 3.6 atm L^2 mol^{-2} and b = 0.05 mol^{-1}L.

(A) 1.23 atm (B) 1.28 atm (C) 2.46 atm (D) 0.64 atm

CSM309

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12. The virial form of van der Waals gas equation is $PV_m = RT\left(1+\frac{B}{V_m}+\frac{C}{V_m^2}+...\right)$. The second virial coefficient of argon gas at 262.5 K is –1 L mol^{-1}. What is the density of argon gas at 262.5 K and 1 atm? Neglect all the terms after second term in the virial forms, under these conditions. (R =0.08 L-atm/K-mol, Ar = 40)

(A) 2.0 g/L (B) 1.905 g/L (C) 1.818 g/L (D) 1.964 g/L

CSM310

Liquefaction; Boyle's Temperature; Graphs of Liquefaction of Real Gas (Andrew's Isotherm of CO_2)

13. The Vander Waals constants for three gases R, S and T are given below.

Gas	a(atm L^2/mol^2)	b(l/mol)
R	5.2	0.420
S	3.8	0.038
T	2.3	0.32

The correct sequence of placing the easily liquefiable gas last and the one difficult to liquefy first is

(A) R, S, T (B) S, T, R (C) T, S, R (D) S, R, T

CSM227

14. The correct order of the values of critical temperature, T_C; Boyle temperature, T_B and inversion temperature, T_i of a real gas is

(A) $T_C < T_B < T_i$ (B) $T_B < T_i < T_C$ (C) $T_B < T_C < T_i$ (D) $T_C < T_i < T_C$

CSM228

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15. One mole of each gases A, B, C and D with van der Waals constant (atm L^2 mol^{-2}) 1.348, 6.823, 4.390 and 2.438, respectively, are kept separately in four different vessels of equal volumes at identical temperature. Their pressures are observed to be P_A, P_B, P_C and P_D, respectively. On the basis of this data alone, the order of pressure of gases is (assume other van der Waals constant to be nearly same for all gases)

(A) $P_A < P_B < P_C < P_D$ (B) $P_B < P_C < P_D < P_A$

(C) $P_B < P_A < P_C < P_D$ (D) $P_C < P_B < P_D < P_A$

CSM303

Critical Temperature; Critical Volume; Critical Pressure

16. For a certain gas obeying van der Waals equation, the critical temperature is 305.5 K and the critical pressure is 48.2 atm. The critical volume is

(A) 0.098 L/mol (B) 0.195 L/mol (C) 0.231 L/mol (D) 0.321 L/mol

CSM225

17. The critical constants P_C and T_C for four gases are as follows.

Gas	Critical temperature	Critical pressure
O_2	153 K	50 bar
Cl_2	417 K	76 bar
H_2	33.9 K	13.5 bar
He	5.12 K	2.2 bar

The gas with the smallest value of van der Waals constant b is

(A) O_2 (B) Cl_2 (C) H_2 (D) He

CSM226

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	A	C	B	A	C	B	C	B	A
Question	11	12	13	14	15	16	17			
Answer	A	A	C	A	B	B	D			

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